FreshCart Gold Layer

Model documentation

WAREHOUSE	BUILT WITH	REFRESH
ClickHouse	**dbt**	**Full rebuild every night** (Airflow)

Business rules used in every model

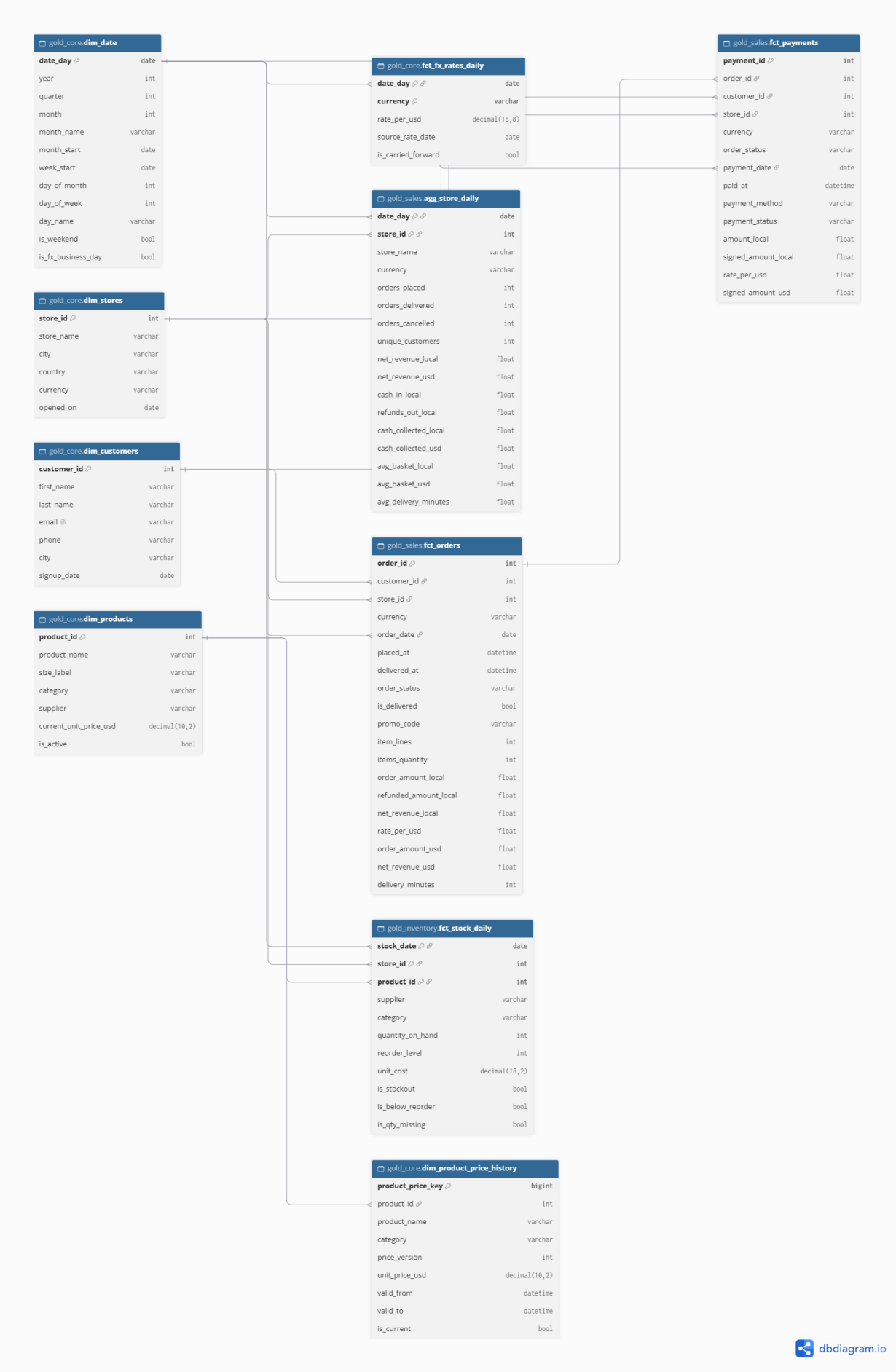
Rule	Definition
Day	Cairo business day: toDate(timestamp, 'Africa/Cairo'). All source timestamps are stored in UTC.
Net revenue	Order lines of delivered orders minus refunds on those orders. Cancelled and open orders = 0.
Cash collected	Payments minus refunds, on the Cairo day the money moved (`paid_at`).
USD conversion	`amount_usd` = `amount_local` / `rate_per_usd` of that day. Friday, Saturday and unpublished days use the last published rate.
Customers	One `customer_id` per person: duplicate accounts merged in silver, QA test accounts removed.
Stockouts	`quantity_on_hand` = 0. N/A quantities and days before a store opened are not stockouts.

Business question → model

Question	Model
Net revenue and cash per store and day, USD and local	`agg_store_daily` (detail: `fct_orders`, `fct_payments`)
Number of orders, average basket, delivery time	`agg_store_daily`, `fct_orders`
Customers (two accounts count once, test accounts excluded)	`dim_customers`, `fct_orders.customer_id`
Stockouts per store, category and supplier	`fct_stock_daily`
Product price history (SCD type 2)	`dim_product_price_history`

Entity-relationship diagram

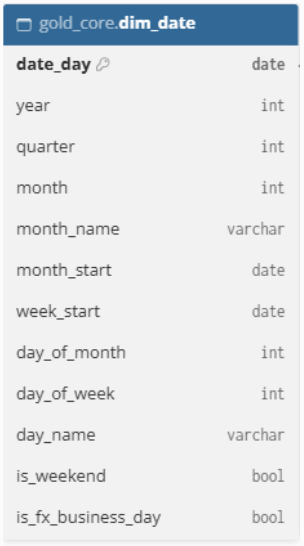
Tables, keys and relationships of the gold layer (gold_core, gold_sales, gold_inventory).



`gold_core` shared dimensions

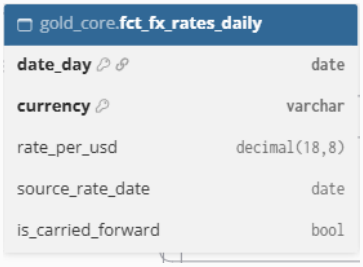
dim_date

TYPE	Dimension
GRAIN	1 row per calendar day (Cairo)
SOURCE	generated calendar (numbers()), from 2023-01-01 (first FX rate) to today; `silver_history` for FX publication days
DESCRIPTION	Calendar used by every daily question. It's generated rather than taken from orders, so days with no orders still exist. Weekend = Friday and Saturday (Egypt / KSA / UAE retail week).
KEY COLUMNS	`date_day` (PK), `year`, `quarter`, `month`, `month_name`, `month_start`, `week_start` (Monday), `day_of_week` (1 = Monday … 7 = Sunday), `day_name`, `is_weekend`, `is_fx_business_day`
TESTS	• `date_day` unique and not null; flags not null



fct_fx_rates_daily

TYPE	Helper fact
GRAIN	1 row per day and currency (EGP, SAR, AED, USD)
SOURCE	`silver_history` (latest usable rate per business day)
DESCRIPTION	Exchange rate (units of currency per 1 USD) for every calendar day. The provider publishes Sunday to Thursday only, so other days carry the last published rate forward (ASOF JOIN). USD = 1 is included so USD amounts can use the same join.
KEY COLUMNS	`date_day`, `currency`, `rate_per_usd`, `source_rate_date` (publication used), `is_carried_forward`
TESTS	• unique (`date_day`, currency); `rate_per_usd` > 0 and not null; currency in EGP / SAR / AED / USD; `date_day` exists in `dim_date`



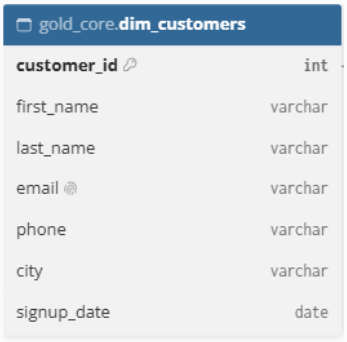
dim_stores

TYPE	Dimension
GRAIN	1 row per store
SOURCE	`silver_stores`
DESCRIPTION	The 7 FreshCart stores with their city, country and local currency. Store 7 (Cairo - Heliopolis) opened on 2026-10-19. All local amounts in the facts are in the store's currency.
KEY COLUMNS	`store_id` (PK), `store_name`, `city`, `country` (EG / SA / AE), `currency` (EGP / SAR / AED), `opened_on`
TESTS	• `store_id` unique and not null; currency not null



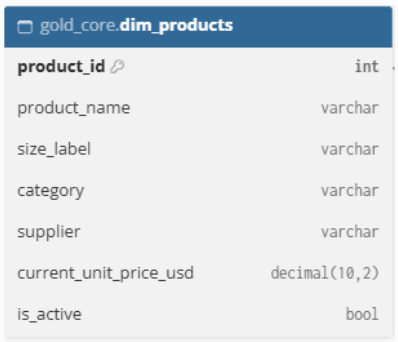
dim_customers

TYPE	Dimension
GRAIN	1 row per person
SOURCE	`silver_customers`
DESCRIPTION	Customers, with one row per real person: duplicate accounts are merged (one account per email, and their orders are remapped in silver), and QA test accounts are excluded. Counting `customer_id` therefore counts people.
KEY COLUMNS	`customer_id` (PK), `first_name`, `last_name`, `email` (unique, lowercase), `phone` (international format), `city`, `signup_date`
TESTS	• `customer_id` unique and not null; email unique



dim_products

TYPE	Dimension (type 1, current values)
GRAIN	1 row per product
SOURCE	`silver_products` + `silver_supplier_stock` (supplier)
DESCRIPTION	Current product attributes, with category and supplier (FRESH: fruits & vegetables, dairy, bakery, meat & fish; GENERAL: everything else). `is_active` is only a catalogue flag: inactive products are still sold, so sales must never be filtered on it. Price history is in `dim_product_price_history`.
KEY COLUMNS	`product_id` (PK), `product_name`, `size_label`, `category`, `supplier`, `current_unit_price_usd`, `is_active`
TESTS	• `product_id` unique and not null; category not null; supplier in FRESH / GENERAL



dim_product_price_history

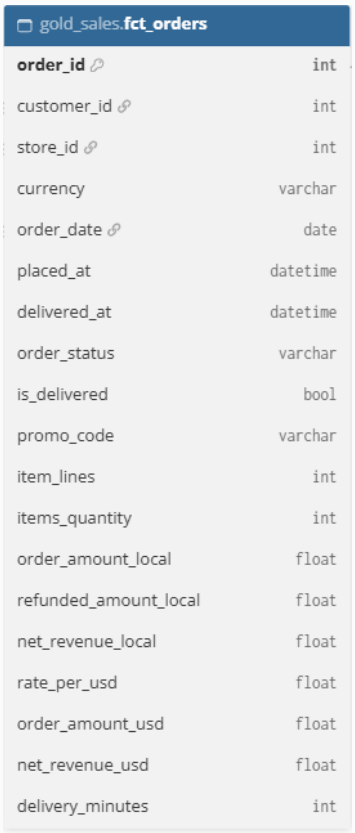
TYPE	Dimension, SCD type 2
GRAIN	1 row per product price version
SOURCE	`silver_product_prices` + `silver_products`
DESCRIPTION	The full catalogue price history in USD since 2023, one row per version with its validity period. The price in effect for an order is the row where `placed_at` >= `valid_from` and (`placed_at` < `valid_to` or `valid_to` is null). Both columns are UTC timestamps, so compare timestamps, not dates.
KEY COLUMNS	`product_price_key` (surrogate key), `product_id`, `product_name`, `category`, `price_version`, `unit_price_usd`, `valid_from`, `valid_to` (NULL = current), `is_current`
TESTS	• `product_price_key` unique and not null; `product_id` exists in `dim_products`; `unit_price_usd` and `valid_from` not null



gold_sales orders, payments and daily store results

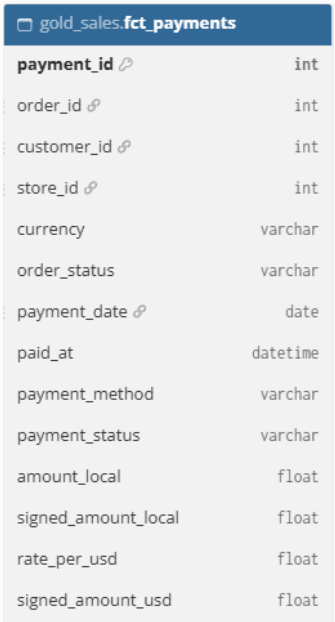
fct_orders

TYPE	Fact
GRAIN	1 row per order
SOURCE	`silver_orders`, `silver_order_items` (summed per order), `silver_payments` (refunds), `silver_stores`, `fct_fx_rates_daily`
DESCRIPTION	Every order with its basket, status, delivery time and net revenue, in local currency and USD (rate of the order day). Answers number of orders, average basket, delivery time and net revenue. Order lines that arrive one night late are picked up by the next nightly rebuild.
KEY COLUMNS	**Keys** `order_id` (PK), `customer_id`, `store_id`, `order_date` **Status** `order_status`, `is_delivered` **Basket** `item_lines`, `items_quantity`, `order_amount_local` / `order_amount_usd` **Revenue** `refunded_amount_local`, `net_revenue_local` / `net_revenue_usd`, `rate_per_usd` **Delivery** `delivery_minutes` (delivered orders only)
TESTS	• `order_id` unique and not null. • Relationships to `dim_customers`, `dim_stores` and `dim_date`. • `order_status` in placed / delivered / cancelled. • `rate_per_usd` not null.



fct_payments

TYPE	Fact
GRAIN	1 row per payment or refund
SOURCE	`silver_payments`, `silver_orders`, `silver_stores`, `fct_fx_rates_daily`
DESCRIPTION	Every movement of money, on the Cairo day it happened. `signed_amount` is positive for payments and negative for refunds, so its sum is cash collected. Refunds count on their own day, whatever the order status. Payments whose order isn't loaded yet appear the next night.
KEY COLUMNS	`payment_id` (PK), `order_id`, `customer_id`, `store_id`, `payment_date`, `payment_method` (card / wallet / cash), `payment_status` (paid / refunded), `amount_local`, `signed_amount_local`, `signed_amount_usd`, `rate_per_usd`
TESTS	• `payment_id` unique and not null; `order_id` exists in `fct_orders`; `payment_status` in paid / refunded; `store_id`, `payment_date` and `rate_per_usd` not null



agg_store_daily

TYPE	Aggregate
GRAIN	1 row per store and Cairo day
SOURCE	`dim_date`, `dim_stores`, `fct_orders`, `fct_payments`
DESCRIPTION	Daily results per store, from the store's first activity (its opening day, or an earlier order) to today. Answers the first two business questions in one table: net revenue and cash collected per store and day, in local currency and USD; number of orders, average basket and average delivery time; unique customers. Days without activity have NULL measures.
KEY COLUMNS	**Keys** `date_day`, `store_id`, `store_name`, `currency` **Orders** `orders_placed`, `orders_delivered`, `orders_cancelled`, `unique_customers` **Revenue** `net_revenue_local` / `net_revenue_usd` **Cash** `cash_in_local`, `refunds_out_local`, `cash_collected_local` / `cash_collected_usd` **Averages** `avg_basket_local` / `avg_basket_usd`, `avg_delivery_minutes`
TESTS	• unique (`store_id`, `date_day`); keys not null; reconciliation tests below

gold_sales.**agg_store_daily**

date_day	date
store_id	int
store_name	varchar
currency	varchar
orders_placed	int
orders_delivered	int
orders_cancelled	int
unique_customers	int
net_revenue_local	float
net_revenue_usd	float
cash_in_local	float
refunds_out_local	float
cash_collected_local	float
cash_collected_usd	float
avg_basket_local	float
avg_basket_usd	float
avg_delivery_minutes	float

gold_inventory stock

fct_stock_daily

TYPE	Fact (periodic snapshot)
GRAIN	1 row per product, store and day
SOURCE	`silver_supplier_stock` (FRESH + both GENERAL file layouts), `silver_products`
DESCRIPTION	Units on the shelf at 22:00 each day, from the two suppliers' end-of-day files. Rows the suppliers reported for days before a store opened are excluded. Answers stockouts per store, category and supplier.
EXAMPLE	`select store_id, category, supplier, countIf(is_stockout) from fct_stock_daily group by store_id, category, supplier`
KEY COLUMNS	`stock_date`, `store_id`, `product_id`, `supplier`, `category`, `quantity_on_hand` (NULL = N/A), `reorder_level`, `unit_cost` (GENERAL from 2026-10-17), `is_stockout`, `is_below_reorder`, `is_qty_missing`
TESTS	• unique (`stock_date`, `store_id`, `product_id`); relationships to `dim_date`, `dim_stores` and `dim_products`; supplier in FRESH / GENERAL; `is_stockout` not null

gold_inventory.**fct_stock_daily**

stock_date	date
store_id	int
product_id	int
supplier	varchar
category	varchar
quantity_on_hand	int
reorder_level	int
unit_cost	decimal(18,2)
is_stockout	bool
is_below_reorder	bool
is_qty_missing	bool

Reconciliation tests `tests/gold/`

Test	Checks	Severity
`agg_store_daily_reconcile_revenue`	Net revenue per store in `agg_store_daily` = `fct_orders` (local and USD)	**error**
`agg_store_daily_reconcile_cash`	Cash collected per store in `agg_store_daily` = `fct_payments`	**error**
`fct_orders_revenue_equals_cash`	For each delivered order, net revenue = cash collected for it (two ways of computing the same number)	**warn**